

Mathematical Formulation of PV Battery Sizing

Energy Optimization Model

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1 Signal definitions (continuous time)

Let:

- $P_{pv}(t)$ = total PV generation (sum of all inverters)
- $P_{net}(t)$ = grid exchange at meter
 - $P_{net}(t) > 0$: import from grid
 - $P_{net}(t) < 0$: export to grid

2 Identification of real consumption

The meter signal does not measure consumption directly but:

$$P_{net}(t) = P_{C,real}(t) - P_{pv}(t) \quad (1)$$

Hence the true consumption is exactly:

$$P_{C,real}(t) = P_{net}(t) + P_{pv}(t) \quad (2)$$

This identity holds pointwise in time and does not require aggregation.

3 Energy balance identity

Define instantaneous decomposition:

$$P_{pv}(t) = P_{C,real}(t) + P_{export}(t) - P_{import}(t) \quad (3)$$

$$P_{export}(t) = \max(-P_{net}(t), 0) \quad (4)$$

$$P_{import}(t) = \max(P_{net}(t), 0) \quad (5)$$

4 Why time resolution matters

Battery optimization is fundamentally path-dependent:

$$\int P_{pv}(t) dt \neq \text{usable self-consumed energy} \quad (6)$$

because the ordering of production and consumption events determines grid exchange.

5 Net energy process

Define the cumulative net energy imbalance:

$$E(t) = \int_0^t (P_{pv}(\tau) - P_{C,real}(\tau)) d\tau \quad (7)$$

Using the meter identity:

$$E(t) = - \int_0^t P_{net}(\tau) d\tau \quad (8)$$

Thus $E(t)$ fully characterizes the system imbalance.

6 Battery as constrained storage process

Let $S(t)$ be the battery state of charge:

$$0 \leq S(t) \leq C \quad (9)$$

with capacity C .

7 Optimal offline battery (perfect foresight)

The optimal storage trajectory is the reflected process:

$$S(t) = \min\left(C, E(t) - \inf_{\tau \leq t} E(\tau)\right) \quad (10)$$

This corresponds to a Skorokhod reflection of $E(t)$ with upper bound C .

8 Physical interpretation

The cumulative process $E(t)$ encodes all mismatch between production and demand.

- $\inf_t E(t)$ defines the deepest deficit state
- $\sup_t E(t)$ defines maximum surplus state

Hence the ideal unconstrained storage requirement is:

$$C_{ideal} = \sup_t E(t) - \inf_t E(t) \quad (11)$$

9 Self-consumption metrics

Define exported energy:

$$E_{exp}^{tot} = \int \max(-P_{net}(t), 0) dt \quad (12)$$

With battery capacity C , residual export becomes:

$$E_{exp}^{res}(C) = \int [\max(-P_{net}(t), 0) - \Delta S_C(t)]^+ dt \quad (13)$$

where $\Delta S_C(t)$ is the charge accepted by the battery.

Self-consumption ratio:

$$\eta(C) = 1 - \frac{E_{exp}^{res}(C)}{E_{exp}^{tot}} \quad (14)$$

10 Role of irradiance model

Let $I(t)$ be irradiance or theoretical yield driver:

$$P_{pv}(t) = \alpha I(t) + \epsilon(t) \quad (15)$$

Then consumption becomes:

$$P_{C,real}(t) = P_{net}(t) + \alpha I(t) + \epsilon(t) \quad (16)$$

Interpretation

- $I(t)$ deterministic physical driver
- $\epsilon(t)$ stochastic PV degradation + noise
- consumption is embedded in residual dynamics via $P_{net}(t)$

11 Discrete formulation (array form)

For samples $i = 1, \dots, N$:

$$E_i = \sum_{k=1}^i (P_{pv,k} - P_{C,k}) \Delta t_k \quad (17)$$

Battery state:

$$S_i = \min\left(C, E_i - \min_{j \leq i} E_j\right) \quad (18)$$

12 Key reduction

Since $P_{C,real}$ is not independently observed:

$$E_i = - \sum_{k=1}^i P_{net,k} \Delta t_k \quad (19)$$

Thus battery sizing depends only on the cumulative meter signal.

Battery sizing is the geometry of the cumulative net energy process $E(t)$.

13 Optimization criteria

1. Physical optimum:

$$C = \sup E - \inf E \quad (20)$$

2. Economic optimum:

$$\arg \max_C (\text{savings}(C) - \text{cost}(C)) \quad (21)$$

3. Marginal saturation point:

$$\frac{d\eta}{dC} \approx 0 \quad (22)$$

14 Final insight

The system reduces to a single stochastic process:

$$E(t) = - \int_0^t P_{net}(\tau) d\tau \quad (23)$$

All PV structure, consumption behavior, and irradiance effects are implicitly encoded in its path.